

UNIT 4: MACHINE LEARNING

TOPIC: DECISION BOUNDARY & COST FUNCTION OPTIMIZATION (SIMPLE EXAMPLES)

1. Decision Boundary (Simple Explanation)

Definition:

A decision boundary is a limit or rule that separates different categories.

Example 1: Pass vs Fail

- Study hours $\geq 5 \rightarrow$ Pass
- Study hours $< 5 \rightarrow$ Fail

👉 Decision Boundary = 5 hours

Example 2: Spam Email

- Spam words $> 3 \rightarrow$ Spam
- Spam words $\leq 3 \rightarrow$ Not Spam

👉 Decision Boundary = 3 words

Example 3: Height Classification

- Height ≥ 170 cm $\rightarrow$ Tall
- Height < 170 cm $\rightarrow$ Short

👉 Decision Boundary = 170 cm

Example 4: Shopping Discount

- Purchase $\geq ₹1000 \rightarrow$ Discount
- Purchase $< ₹1000 \rightarrow$ No Discount

👉 Decision Boundary = ₹1000

Key Idea:

A decision boundary is simply a **cutoff value** that helps separate categories.

2. Cost Function Optimization (Simple Explanation)

Definition:

A cost function measures how wrong a prediction is.
Optimization means reducing that error step by step.

Example 1: Guessing Marks

- Actual marks = 80
- Predicted = 70 $\rightarrow$ Error = 10

Next guess:

- Predicted = 78 $\rightarrow$ Error = 2

👉 Error is reduced $\rightarrow$ Optimization

Example 2: Throwing a Ball at Target

- First throw $\rightarrow$ far from target
- Second throw $\rightarrow$ closer
- Third throw $\rightarrow$ very close

👉 Distance from target = Cost

👉 Reducing distance = Optimization

Example 3: Delivery Time Prediction

- Predicted time = 30 min
- Actual time = 40 min $\rightarrow$ Error = 10

Next prediction:

- 38 min $\rightarrow$ Error = 2

👉 Better prediction = Lower cost

Example 4: Temperature Prediction

- Actual = 35°C
- Predicted = 30°C $\rightarrow$ Error = 5

Next prediction:

- $34^{\circ}\text{C} \rightarrow \text{Error} = 1$

👉 Smaller error = Better model

Example 5: Weight Loss Goal

- Target weight = 70 kg
- Current = 80 kg $\rightarrow$ Difference = 10

Progress:

- $78 \rightarrow 75 \rightarrow 72 \rightarrow 70$

👉 Difference reduces $\rightarrow$ Optimization complete

Final Understanding

Decision Boundary:

A rule or limit that separates categories.

Cost Function Optimization:

A process of reducing mistakes step by step to improve accuracy.

1. Decision Boundary

Definition:

A decision boundary is a line, curve, or surface that separates different classes in a dataset.

- In 2D $\rightarrow$ line/curve
- In 3D $\rightarrow$ plane
- In higher dimensions $\rightarrow$ hyperplane

Mathematical Form (Linear):

$$w_1x_1 + w_2x_2 + b = 0$$

Where:

- $w_1, w_2 \rightarrow$ weights
- $b \rightarrow$ bias

Logistic Regression Decision Boundary:

$$P(y=1) = 1 / (1 + e^{-(w^T x + b)})$$

Boundary occurs when:

$$w^T x + b = 0$$

Types of Decision Boundaries:

1. Linear Boundary
 - Straight line
 - Used in Logistic Regression, Linear SVM
2. Non-Linear Boundary
 - Curved
 - Used in Decision Trees, Neural Networks

Example:

x_1	x_2	Class
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2	3	0
---	---	---

4	5	0
---	---	---

6	7	1
---	---	---

8	9	1
---	---	---

A line like:

$x_2 = x_1 + 1$ separates the classes.

Key Points:

- Separates classification regions
- Depends on model parameters
- Learned during training

2. Cost Function Optimization

Definition:

A cost function measures how incorrect the model predictions are.

Goal:

Minimize the cost function

$J(\theta) \rightarrow \text{minimum}$

Mean Squared Error (MSE):

$$J = (1/n) \sum (y_{\text{pred}} - y_{\text{actual}})^2$$

Example:

Actual: 10, 20

Predicted: 8, 22

$$\text{MSE} = [(10-8)^2 + (20-22)^2]/2 = 4$$

Log Loss (Binary Cross Entropy):

$$J = -1/n \sum [y \log(p) + (1-y) \log(1-p)]$$

Used in classification problems.

Gradient Descent

Formula:

$$\theta = \theta - \alpha (dJ/d\theta)$$

Where:

- α = learning rate
- $dJ/d\theta$ = gradient

Types of Gradient Descent:

1. Batch Gradient Descent
 - Uses full dataset
 - Accurate but slow
2. Stochastic Gradient Descent
 - Uses one data point
 - Fast but noisy
3. Mini-batch Gradient Descent
 - Balanced approach

Example:

$$J(w) = (w - 5)^2$$

Minimum occurs at: $w = 5$

Learning Rate (α)

- Small $\alpha \rightarrow$ slow learning
- Large $\alpha \rightarrow$ unstable

Challenges in Optimization

- Local minima
- Saddle points
- Overfitting

Regularization

$$J = \text{Loss} + \lambda \sum w^2$$

Prevents overfitting by penalizing large weights.

Connection Between Topics

- Cost function adjusts parameters
- Parameters define decision boundary

Therefore:

Minimizing cost $\rightarrow$ Better decision boundary

Summary

- Decision boundary separates classes
- Cost function measures prediction error
- Gradient descent minimizes cost
- Optimization improves model accuracy